

Department of Electronics and Communication Engineering

III C-212

Data Communication & Networking

Max. Marks: 15

Time: 50 Minutes

Note: All Questions are compulsory

- Q1. Compare and contrast the Go-Back-N-ARQ Protocol with Selective-Repeat ARQ. (3)
- Q2. Given the dataword 1010011110 and the divisor 10111, a) Show the generation of the codeword at the sender site (using binary division). b) Show the checking of the codeword at the receiver site (assume no error). (4)
- Q3. Find the minimum Hamming distance for the following cases: a) Detection of two errors. b) Correction of two errors. c) Detection of 3 errors or correction of 2 errors. d) Detection of 6 errors or correction of 2 errors. (4)
- Q4. In a CDMA/CD network with a data rate of 10 Mbps, the maximum distance between any station pair is found to be 2500 m for the correct operation of the collision detection process. What should be the maximum distance if we increase the data rate to 100 Mbps? To 1 Gbps? To 10 Gbps? (4)